

ESAWAAI – Ready Legacy Dataset Report

CLS-ENV-NT-24-0452

1.0

Open/Public/Público

CHRONOLOGY ISSUES/HISTORIQUE DES VERSIONS

Issue/ Version	Date	Object/Objet	Written by/ Rédigé par	Checked by / Vérifié par	Approved by/ Approuvé par
1.0	12/08/2024	Creation	Amine Benchaabane	Romain Husson	

DISTRIBUTION/LISTE DE DIFFUSION

Company/Organisme	Means of distribution/ Format de diffusion	Names/Destinataires
European Space Agency	electronic	

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1 Introduction

This document serves as the Ready Legacy Dataset Report (DR) for the ESAWAAI project, a key component aimed at providing a comprehensive understanding of the dataset, its relevance, and its applications. It is structured into four main sections to guide users through the key aspects of the project and its data:

1. **Project Objectives**
2. **Legacy Dataset Description**
3. **ESAWAAI Product Description**
4. **Conclusion**

The report begins by outlining the project objectives, providing context for the development and purpose of the ESAWAAI project. It details the identification of critical Synthetic Aperture Radar (SAR) observables that are essential to the project's wind retrieval approach, as well as the reference dataset used for target analysis. Further, it explains the process and rationale behind the selection of specific regions of interest and time periods for data collection, ensuring that these choices are made to maximize the quality and relevance of the legacy dataset.

The dataset is structured to ensure that it captures the necessary information for accurate wind field estimation and other related applications. Each of the SAR observables included in the dataset is considered with respect to its limitations and potential to improve the wind field retrieval process. This document will be continuously updated by the data provider to reflect the latest developments and ensure the dataset remains relevant and useful for external users.

2 Project objectives

The ESAWAAI project is dedicated to advancing the retrieval of sea surface wind fields using a broad range of observables extracted from multi-polarization single-look complex (SLC) SAR data. By leveraging cutting-edge technologies, the project seeks to build a unique and comprehensive legacy dataset that will serve as a foundation for wind retrieval applications. This dataset will not only be critical for improving the accuracy of wind field estimation but also for the development and implementation of Deep Learning (DL) models. These models will incorporate a Bayesian scheme, which effectively integrates physics-based models, to refine the retrieval process further and enhance the reliability of the wind field estimations.

Sentinel-1 SAR sensors are central to the ESAWAAI project due to their ability to provide a wealth of observables that extend beyond the traditional single-polarization sigma0. These additional variables, including cross-polarization sigma0, polarimetric parameters such as Co-Cross Polarimetric Coherence, sub-looks estimates, geophysical Doppler, and the Imaginary part of MeAn Cross-Spectra (IMACS), offer valuable insights into wind field dynamics. While each of these observables provides unique information, they also present challenges, particularly when it comes to fully characterizing wind fields across diverse geometries and environmental conditions. Understanding the limitations of these variables is crucial for improving wind field retrieval accuracy. Moreover, by combining and using these observables in a complementary manner, the project aims to address these limitations, enabling more precise wind field estimations and enhanced uncertainty quantification—independent of any reliance on Numerical Weather Prediction (NWP) models.

In its initial phase, the ESAWAAI project is focused on the creation of this unprecedented legacy dataset, drawing upon the expertise of professionals in SAR processing, Artificial Intelligence (AI), and explainable AI (XAI) technologies. This collaborative approach will ensure that the dataset is both comprehensive and adaptable to the various downstream Earth Observation (EO) applications for which it is intended. The development of this dataset is a crucial step in enabling more accurate and reliable retrieval of sea surface wind fields, which will have significant implications for multiple domains.

The resulting sea surface wind fields will provide essential input for a variety of applications, particularly in meteorology and wind resource assessment. With more accurate wind field retrieval, the project is expected to improve weather forecasting and facilitate better planning for renewable energy production from wind resources. Furthermore, one of the key objectives of the ESAWAAI project is to gain a deeper understanding of how the different SAR observables contribute to wind field estimation. This includes understanding their dependencies on varying metocean conditions—such as the interaction between meteorological and oceanographic factors—and how the observing geometry influences the effectiveness of the wind retrieval models. By achieving this, the project will not only enhance the technical understanding of SAR-derived wind fields but will also make significant strides in improving the application of SAR data in a wide range of practical scenarios.

3 Legacy dataset description

3.1 SAR features

3.1.1 Generalities

Sentinel-1 is a SAR constellation of polar-orbiting satellites at an altitude of 693 km, designed for high-resolution, all-weather imaging. Sentinel-1A has been operational since 2014 with a 12-day repeat cycle, providing critical data despite the loss of Sentinel-1B in December 2021 due to an anomaly. For the ESAWAAI project, the focus is on multi-polarization Single Look Complex (SLC) Level 1 products from Sentinel-1A, specifically using Interferometric Wide (IW) swaths. These products offer detailed radar data, crucial for retrieving sea surface wind fields and developing the project's legacy dataset and Deep Learning models.

The Sentinel-1 L1 Single Look Complex (SLC) products in Interferometric Wide (IW) mode are processed through two intermediate levels, L1B and L1C by IFREMER using the XSARSLC processor. This processor was developed as part of the SARWAVES project, which provides an advanced framework for processing SAR data. More detailed information about these products and the processing levels can be found in the product documentation provided by IFREMER: [Level-1B SAR Ifremer Product Description — **xsarslc**](#): SAR processor for ocean Level-1 SLC -> Level-1B XSP documentation. This processing chain ensures that the raw data is converted into a format suitable for further analysis and application, particularly in the context of sea surface wind field retrieval within the ESAWAAI project.

The Sentinel-1 L1 SLC/IW products used in this project are sourced from the Copernicus Data Space Ecosystem (CDSE) for the identified Regions of Interest (ROIs), covering the period from June 2019 to March 2023. These products are key to the ESAWAAI project's objective of retrieving sea surface wind fields, as they offer high-resolution radar data suitable for detailed analysis. However, certain SAR features, especially those that require finer spatial resolution, cannot be directly retrieved from the SLC/IW products alone, particularly when dealing with high-resolution data typically at a 1 km scale. As a result, additional data from Level 2 Ocean ([L2 OCN](#)) products—processed from Level 1 Ground Range Detected ([L1 GRD](#)) SAR products—are also incorporated into the project.

To facilitate the use of these features in the context of wind field retrieval, a series of projections are applied to accurately map the data onto the appropriate L1B grids. This ensures that the various SAR features, which may come from different processing levels or resolutions, are aligned and compatible for further analysis. These L1B grids, which are crucial for the project's data processing pipeline, will be described in greater detail in the following sections of the document. By integrating data from multiple sources and applying sophisticated projection techniques, the project aims to enhance the accuracy of the sea surface wind field retrieval while addressing the inherent challenges posed by different data types and formats.

The SAR features in the ESAWAAI project are classified based on their contribution to sea surface wind field retrieval, providing key information on ocean surface characteristics like roughness, wind direction, and wave patterns. These features help improve the accuracy of wind field estimation models, with further details to be provided in subsequent sections.:

- **Radiometric features:**

The ESAWAAI project utilizes two key radiometric features: the Co-polarized Normalized Radar Cross Section (**co-pol NRCS**) [Stoffelen2017] [Yiru2020], which includes VV or HH polarizations, and the Cross-polarized Normalized Radar Cross Section (**cross-pol NRCS**) [Mouche2020], which includes HV or VH polarizations. The co-pol NRCS helps analyze surface roughness and wind patterns, while the cross-pol NRCS provides additional insights into wind-wave interactions. These features, used together, enhance the accuracy of sea surface wind retrieval from SAR data.

- **Polarimetric features:**

The Co-Cross-Polarization Coherence (**CCPC**) parameter, as described in [Longepe2019], measures the coherence between co-polarized and cross-polarized radar signals, providing insights into surface roughness and wind-wave interactions. It enhances the accuracy of sea surface wind field retrieval by

distinguishing between different scattering mechanisms, complementing other SAR features like co-pol and cross-pol NRCS. This contributes to more precise wind estimations in the ESAWAAI project.

- **Doppler features:**

The Ocean Surface Radial Velocity (**RVL**) is estimated based on the Geophysical Doppler Centroid Anomaly, which captures the Doppler shift caused by the movement of the ocean surface. This shift, when weighted by the Normalized Radar Cross Section (NRCS), provides a measure of the radial velocity of the ocean surface. Essentially, the RVL reflects the speed and direction of surface motion, influenced by wind and ocean currents. This parameter has been similarly retrieved from Real Aperture Radars (RAR), as demonstrated in studies such as [Moiseev2020]. The RVL offers valuable information for understanding ocean dynamics, especially in the context of wind field retrieval, by helping to characterize the motion of the ocean surface more precisely.

- **Spectral features:**

- The **Imaginary part of MeAn Cross-Spectra (IMACS)**, as described in [Li2019], is derived from filtering - SAR cross-spectra within the 15–20-meter range-detected wavelengths. IMACS is sensitive to increasing wind speeds and provides insights into the directional signatures of intermediate ocean waves, which are influenced by local wind fields. This helps reduce wind direction ambiguity in high-resolution wind field inversions. Similarly to the Doppler anomaly parameter, IMACS aids in accurately estimating wind direction.
- The **Azimuth Cut-off (Az_{off})** is closely related to wind speed and is estimated from the cross-spectra.

- **Textural features:**

The **Approximate Wind Direction ($D_{streaks}$)** is a parameter used to estimate the wind direction based on the presence of wind streaks observed in the SAR co- and cross-polarized sigma0 textures. Wind streaks are surface features caused by the interaction of wind with the ocean, which can be detected through the changes in radar backscatter observed in these textures. By analyzing these textures, particularly in the co- and cross-polarized channels, the approximate wind direction can be derived. The accuracy of this wind direction estimation is influenced by the intensity and orientation of the wind streaks, as well as the quality of the SAR data. Studies such as [Zanchetta2020] and [Husson2021] highlight the potential of this method for approximating wind direction in SAR imagery, providing valuable information for wind field retrieval, especially in cases where direct measurements may be challenging.

To improve wind field retrieval from SAR data, effective pre-processing is necessary to eliminate signatures unrelated to wind-driven backscattering mechanisms. This includes rejecting speckle noise, land contamination, rain or weather effects, rough ocean surface features, and non-wind-driven phenomena. By filtering out these irrelevant signatures, the accuracy and reliability of the wind field estimates can be enhanced:

- In the context of analyzing data from northern latitudes, incorporating **Sea Ice information** derived from SAR data is crucial to ensure that the AI models are not trained on non-free sea surfaces. Sea ice can significantly alter the radar backscatter, leading to inaccuracies in wind field retrieval. Therefore, excluding regions with sea ice is essential for improving model performance. The **AUTO-ICE challenge**, which provides Sea Ice information, could be an ideal source to fulfill this requirement. By leveraging data from this challenge (available at: <https://platform.ai4eo.eu/auto-ice>), the ESAWAAI project can accurately identify and filter out sea ice areas, ensuring that the models are trained on only free sea surfaces, thereby enhancing the reliability of wind field estimations in these regions.
- **Rain segmentation from SAR** data is important for excluding rain-affected areas from the wind retrieval process. By identifying and filtering out rain-induced distortions, as outlined in [Colin2024], the ESAWAAI project ensures that only wind-driven features are considered, improving the accuracy of wind field estimations.
- **Anthropogenic structures**, such as ships and wind farms, must be filtered out from SAR measurements to prevent bias in the SAR observables. These structures can distort the estimation of various parameters,

including co-pol NRCS, cross-pol NRCS, and the RVL. To address this, methodologies such as CFAR (Constant False Alarm Rate) algorithms are typically applied to identify and exclude these bright signatures from the data. These algorithms, as discussed in [Hajduch2008] and [Trunk2008], help to maintain the accuracy of the wind field retrieval process by ensuring that anthropogenic features do not interfere with the SAR measurements.

3.1.2 SAR SLC IW grids

SAR observables, derived from SLC products, can be considered from different geographical coverages, each with its own advantages and disadvantages. These coverages are primarily determined by the type of SAR portions being utilized:

1. **Interburst Coverage:** This type uses SAR portions located in the overlapping regions between two successive bursts, achieved by combining the looks from both bursts. These portions cover a smaller azimuthal region (less than 3 km) and are not spatially continuous, as illustrated in Figure 1.
2. **Intraburst Coverage:** This type utilizes SAR portions within a single burst, employing sublooks from within the same burst. These portions cover a larger azimuthal region (up to 17.5 km) and are spatially continuous, as depicted in Figure 1.

Each coverage type provides distinct advantages, with interburst being more limited in spatial continuity and range, while intraburst offers larger, continuous coverage. The selection between these options depends on the specific needs of the ESAWAAI project.

In the L1B SAR products, the intersection zone between bursts is systematically divided into small areas called "Tiles," where all the relevant SAR variables are estimated. These Tiles also serve as regions of interest for the projection of external and reference data.

While the spatial discontinuity between interburst and intraburst regions may seem like a disadvantage, certain observables, such as the IMACS parameter, may exhibit better wind dependency in the interburst regions. This is because the time gap between SAR looks from two bursts (~2 seconds) is significantly longer than the time between sublooks within a single burst (~0.05 seconds), which is crucial for estimating parameters like IMACS.

As a result, the various SAR-derived variables will be provided for both geometries—covering pixels in both inter- and intraburst regions—ensuring comprehensive data for wind field estimation.

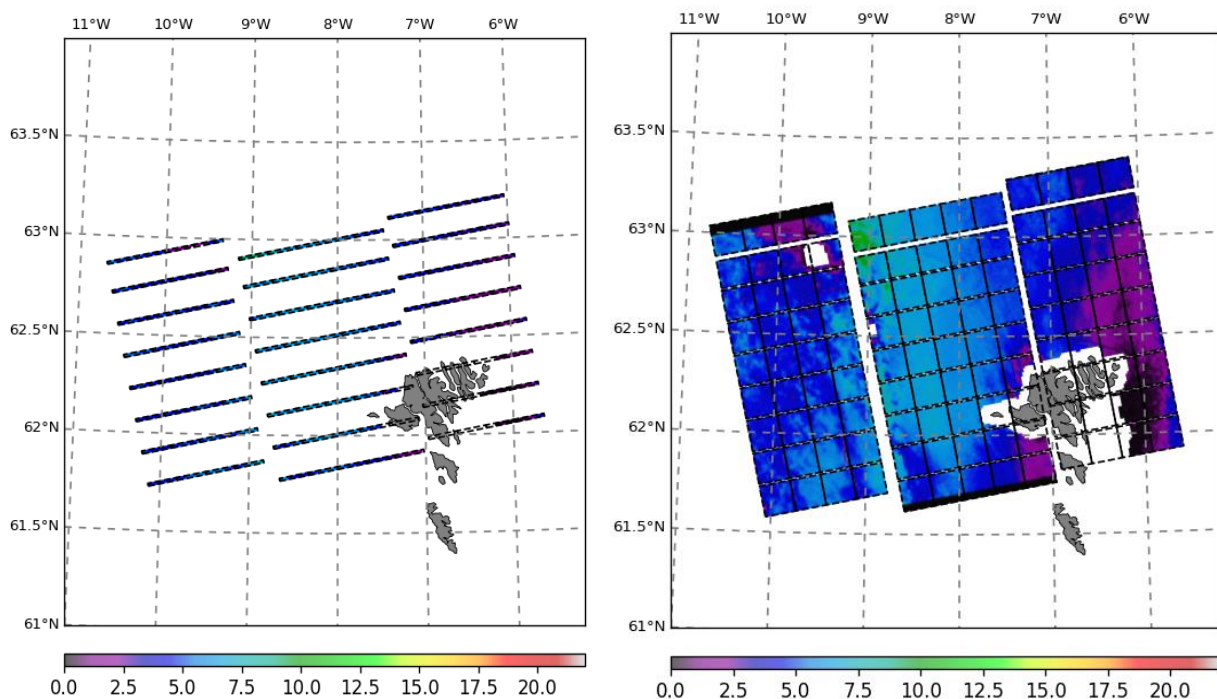


Figure 1: SAR wind speed (m/s) from L2 OCN projected in interburst (left) and intraburst (right) grids of L1C product (2021-01-02T18:08:29 UTC).

It is essential to highlight that, for consistency and accuracy, all wind-related variables—such as wind speed and direction—are systematically transformed into their respective U (east-west) and V (north-south) components. This step is crucial before projecting the data onto the L1B product grids. The conversion to U and V components not only standardizes the wind information but also facilitates more precise and meaningful comparisons across different regions and time periods. By using this method, the data becomes more compatible with other geophysical models and enables better integration with downstream applications, enhancing the overall utility of the wind retrieval process within the ESAWAAI project.

3.2 Reference data

The ESAWAAI project uses two key reference datasets: **NORA3**, which primarily covers the North Sea (Figure 2), and **CERRA**, which spans a larger area over the Mediterranean Sea (Figure 2). The rationale behind choosing these datasets lies in their geographical diversity and the differing wind regimes they represent. The North Sea and Mediterranean Sea each have distinct meteorological and oceanographic conditions, making them ideal for testing the generalization capabilities of the trained models. By evaluating the model's ability to accurately retrieve wind fields in these two regions, the ESAWAAI project aims to ensure that the models are not only effective within their training domain but also robust enough to perform well in areas outside the training scope. This helps demonstrate the versatility and reliability of the wind retrieval models in varying environmental conditions, supporting their broader application in wind field estimation.



Figure 2: Regions of Interest for ESAWAAI project: the North Sea (in red) fully covered by NORA3 data; the Mediterranean Sea (in orange) covered by CERRA data.

3.2.1 NORA3 data

The NORA3 dataset is a high-resolution atmospheric dynamic downscaling of the ERA5 reanalysis data provided by the European Centre for Medium-Range Weather Forecasts (ECMWF). With a 3 km spatial resolution, NORA3 offers a detailed representation of atmospheric conditions, particularly wind fields, over the northern part of the Atlantic Ocean. This high resolution is achieved using the HARMONIE-AROME (H-A) model, a state-of-the-art Numerical Weather Prediction (NWP) system developed and used by numerous European weather forecasting and research institutions [Bengtsson2017]. The H-A model integrates real-time observational data to simulate weather conditions at a finer spatial scale.

NORA3's domain covers a significant portion of the northern Atlantic, including key areas like the North Sea, with a horizontal resolution of 3×3 km. The dataset benefits from a fine atmospheric mesh with at least 30

vertical layers, allowing for a highly granular representation of atmospheric dynamics. This level of detail provides valuable insights into wind behavior at different altitudes, offering an improved understanding of wind patterns and conditions in the region.

In terms of accuracy, NORA3 is generally considered reliable, with wind speed estimates typically being about 5% (or 0.5 m/s) lower than observed values, as highlighted by [Solbrekke2021]. This slight underestimation does not significantly impact the dataset's utility in wind field retrieval, making it an excellent choice for the ESAWAAI project.

For the ESAWAAI project, NORA3 serves as the reference dataset for the North Sea region, providing essential wind speed and direction data. The dataset is available in both **V1.0** and **V2.0** versions of the legacy dataset, which are aligned with the Sentinel-1 SLC/IW grids. These versions of the dataset will be integral to the project's model training and validation processes. By comparing the SAR-derived wind retrievals with the NORA3 data, the ESAWAAI project aims to improve the accuracy and reliability of its wind field estimation models, ensuring they can generalize across varying geographic regions and meteorological conditions.

3.2.2 CERRA data

CERRA (Copernicus European Regional Re-Analysis) is a high-resolution reanalysis system designed to provide pan-European climate data with a horizontal resolution of 5.5 km, which is forced by the global ERA5 reanalysis dataset. CERRA generates essential climate variables using the UERRA-HARMONIE (a 3-dimensional variational data assimilation system) and MESCAN-SURFEX (a complementary surface analysis system).

The CERRA service operates in two phases. In the first phase, the system, developed under the FP7 UERRA project, was used to update the regional reanalysis (RRA) for Europe, initially produced up to December 2015, with the most recent operational update available for July 2019. The data from the UERRA (Uncertainties in Ensembles of Regional Re-analyses) project is available for Europe through the Copernicus Data Store (CDS), covering the period from January 1961 to July 2019. In the second phase, CERRA transitioned to using an updated limited-area weather forecast model and data assimilation methods to produce a more up-to-date regional reanalysis. The production of CERRA data began in early 2020 and continues to be updated regularly.

For the ESAWAAI project, CERRA serves as the reference dataset for the Mediterranean Sea region. It is available in version **V2.0** of the legacy dataset, which is aligned with Sentinel-1 SLC/IW grids. This high-resolution dataset is crucial for wind field retrieval over the Mediterranean region, helping to validate the accuracy of wind retrieval models and ensuring the robustness of the models across various geographical regions.

4 ESAWAAI product definition

4.1 Product structure

Each product in the ESAWAAI legacy dataset is stored using the **SAFE (Standard Archive and Format for Europe)** directory structure, which is the convention inherited from the **ESA Sentinel-1 Level 1 official products**. This structure is designed to ensure compatibility and standardization across different satellite data products. However, for the ESAWAAI project, the “product type” acronym used in the original Sentinel-1 format has been replaced by the project name “**ESAWAAI**” to distinguish the products associated with this specific study (Figure 3). This modification ensures that all the dataset products are easily identifiable and aligned with the objectives of the ESAWAAI project (Figure 3).

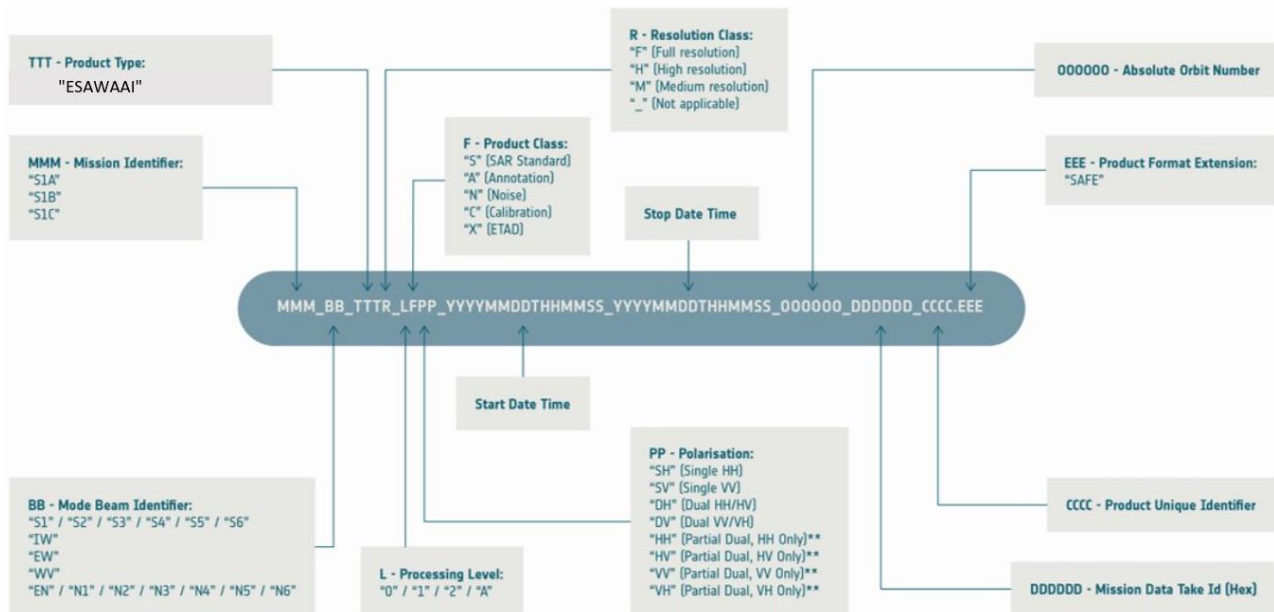


Figure 3: ESAWAAI product name (inherited from [ESA Sentinel-1 level 1 official product](#)).

- **Example of ESAWAAI product name (.SAFE):**

- `S1B_IW_ESAWAAI__1SDV_20210930T064517_20210930T064544_028924_03739E_0D86_B07.SAFE`

Each SAFE product directory in the ESAWAAI legacy dataset contains three netCDF files, with one file per polarization (VV/VH) for each sub-swath. This structure follows the official ESA Sentinel-1 Level 1 product storage convention, ensuring consistency with standard practices. The only modification made is the replacement of the product type acronym “**ttt**” with “**esawaai**,” which uniquely identifies the products associated with the ESAWAAI project (Figure 4). This minor adjustment ensures that the dataset remains compliant with ESA standards while being distinctly tied to the ESAWAAI study (Figure 4).

mmm-sss-ttt-pp-yyyyymmddthhmmss-yyyyymmddthhmmss-000000-ddddddd-nnn.eee

Figure 4 : ESAWAAI sub-swath nomenclature (inherited from [ESA Sentinel-1 level 1 official product](#)).

- **Example of ESAWAAI .SAFE content:**

- `s1b-iw1-esawaai-1sdv-20210930t064519-20210930t064544-028924-03739e-b07.nc`
- `s1b-iw2-esawaai-1sdv-20210930t064519-20210930t064544-028924-03739e-b07.nc`
- `s1b-iw2-esawaai-1sdv-20210930t064519-20210930t064544-028924-03739e-b07.nc`

Each sub-swath netCDF file in the ESAWAAI legacy dataset is organized into two distinct groups: “**intraburst**” and “**interburst**”. These groups correspond to the two grids used for the SLC/IW products as previously discussed. The “intraburst” group contains data from a single burst, providing spatially continuous measurements, while the “interburst” group includes data from the overlapping regions between two successive bursts, which are spatially

discontinuous. This organization ensures that both grid types are properly represented and allows for a comprehensive analysis of SAR data across different spatial configurations.

4.2 Product dimensions

In the ESAWAAI legacy dataset, each sub-swath netCDF file is organized into two groups: "intraburst" and "interburst", corresponding to the two grids used for SLC/IW products. The useful dimensions of each netCDF file, regardless of the grid type, are detailed in Table 1, which outlines the size and structure of the data for each sub-swath, ensuring clarity for further analysis and processing Table 1.

	Dimension	Definition	Useful for ESAWAAI
1	tile_line	The number of Tiles along the azimuth direction (typically One line by burst along the azimuth)	
2	tile_sample	The number of tiles along the range direction	
3	c_sample	The Tile corners along the range direction	
4	c_line	The Tile corners along the azimuth direction	
5	nr_px_line_1km	The number of pixels inside the tile over the azimuth direction at 1 km of resolution	
6	nr_px_sample_1km	The number of pixels inside the tile over the range direction at 1 km of resolution	YES
7	nr_px_line_3km	Number of pixels inside the tile over the azimuth direction at 3 km of resolution (only for intraburst grid)	
8	nr_px_sample_3km	Number of pixels inside the tile over the range direction at 3 km of resolution (only for intraburst grid)	
9	pol	Number of SAR polarizations	
10	Bt_thresh	lower edge of bright target to background amplitude ratio	
11	Lambda_range_max_macs	maximum wavelength bound for MACS estimation	
12	k_gp	Gegenbauer polynoms dimension	No
13	Phi_hf	Harmonic functions dimension (odd number)	(Related to wave tasks)
14	freq_line	Wavenumber vector over range axis	
15	Freq_sample	Wavenumber vector over the azimuth axis	

Table 1 : ESAWAAI netCDF dimensions

4.3 Product resolutions

The ESAWAAI legacy dataset contains radar and geophysical variables related to waves, wind, and Doppler, derived from L1B products at interburst and intraburst resolutions. It also includes reference data projected at different resolutions: 3 km per pixel for the intraburst grid and 1 km per pixel for the interburst grid, given the larger tile size of approximately 3 km in azimuth for the interburst grid. Additionally, other geophysical and atmospheric variables from numerical models are provided at these respective resolutions.

The dataset also incorporates SAR wind speed (used as a baseline wind field), SAR platform heading, and incidence angles at 1 km/pixel, extracted from L2 OCN products. The rain rate, estimated from L1 GRD products, is included at 1 km/pixel resolution as well.

If a variable is projected at a specific resolution, this is clearly stated in its name. If no resolution is mentioned, it indicates that the variable is projected at either the intraburst or interburst tile resolution, depending on the specific netCDF group being accessed. Example:

- **WS10_u**: NORA3 u component projected at tile size (intra or interburst) (Figure 5)
- **WS10_u_3km**: NORA3 U wind component at 3km/pixel (Figure 5)
- **WS10_u_1km**: NORA3 U wind component projected at 1km/pixel (Figure 5)

For each resolution generated, the two corresponding latitude and longitude matrix are systematically generated as the following example:

- **latitude_3km**: latitudes table for variables projected at 3km/pixel
- **latitude_1km**: latitudes table for variables projected at 1km/pixel
- **latitude**: latitudes table for variables at Tile resolution (intra/interburst)

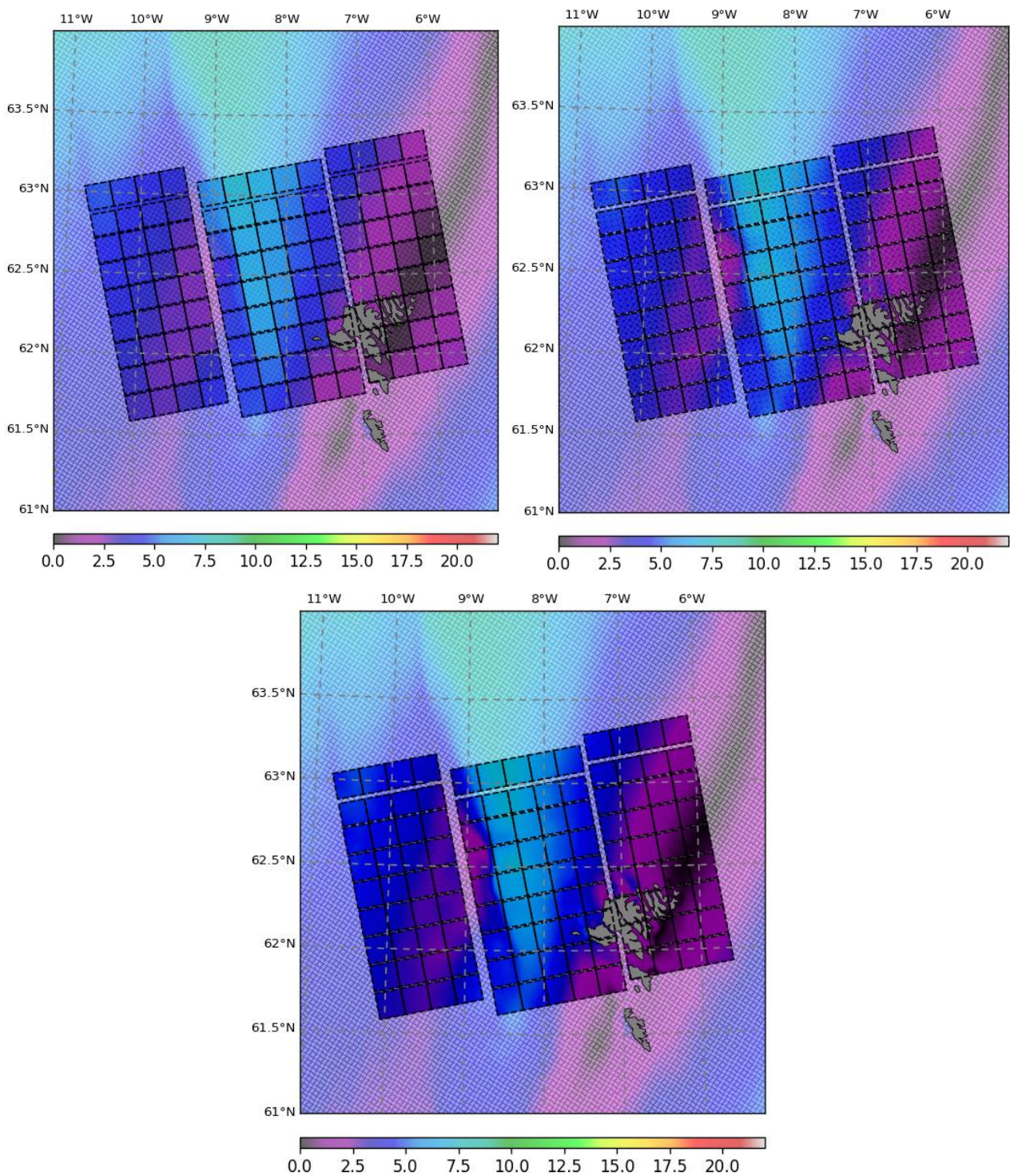


Figure 5 : NORAD wind speed (m/s) projected in intraburst grid at Tile resolution (top left), 3km/pixel (top right) and 1km/pixel (bottom) for SAR acquisition at 2021-01-02T08:08:29. The background map represents the raw NORAD wind speed (3km/pixel).

4.4 Product main variables

In addition to the core variables provided directly in the L1C product (such as wind, waves, and Doppler), the key variables included in the ESAWAAI product are listed in Table 2. Each of these variables is accompanied by its respective attributes within the netCDF file, offering detailed descriptions for better understanding and use.

Feature & Target	Grid	Azimuth resolution	Range Resolution	Name in the legacy dataset
SAR Co(Cross)-pol NRCS*	IntraBurst	~17 km	~17 km	sigma0_filt (provided only for co-pol)
		1 km	1 km	owiNrcs_1km, owiNrcsNeszCorr_1km while the dimension: pol = 0 for co-pol pol = 1 for cross-pol
		50 or 100 m	50 or 100 m	owiNrcs_100m, owiNrcsNeszCorr_100m while the dimension: pol = 0 for co-pol pol = 1 for cross-pol
	InterBurst	~2 km	~17 km	sigma0_filt (only for co-pol)
		1 km	1 km	owiNrcs_1km, owiNrcsNeszCorr_1km while the dimension: pol = 0 for co-pol pol = 1 for cross-pol
SAR Incidence Angle*	IntraBurst	~17km	~17km	incidence
		1km	1km	owiIncidenceAngle_1km
	InterBurst	~2 km	~17 km	incidence
		1km	1km	owiIncidenceAngle_1km
SAR Heading*	IntraBurst & InterBurst	1km	1km	owiHeading_1km
SAR Azimuth Cutoff	IntraBurst	~17km	~17km	azimuth_cutoff
	InterBurst	~2 km	~17 km	
SAR Wind Speed*	IntraBurst & InterBurst	1km	1km	owiWind_u_1km: u component owiWind_v_1km: v component
Retrieved SAR mask*	IntraBurst & InterBurst	1km	1km	owiMask_1km
Percentage of bright targets pixels detected in cell*	IntraBurst & InterBurst	1km	1km	owiPBright_1km
Rain Rate from SAR**	IntraBurst & InterBurst	1km	1km	opr_rr_reg_1km
MACS	IntraBurst	~17km	~17km	macs_Re (Real part) and macs_Im (Imaginary part)
	InterBurst	~2 km	~2 km	
CCPC (Not calibrated)	IntraBurst	~17km	~17km	CCPC_filt_Re (Real Part) and CCPR_filt_Im (Imaginary part)
CCPC (Not calibrated)	InterBurst	~2 km	~2 km	
Wind Streaks	***	***	***	***
NORA3 wind speed	IntraBurst	~17km	~17km	WS10_u: u component WS10_v: v component
		3km	3km	WS10_u_3km: u component WS10_v_3km: v component
		1km	1km	WS10_u_1km: u component WS10_v_1km: v component
	InterBurst	~2 km	~17 km	WS10_u: u component WS10_v: v component

		1km	1km	WS10_u_1km: u component WS10_v_1km: v component
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*Extracted from L2 OCN SAR product

** Estimated from L1 GRDH SAR product

*** Not yet available

Table 2 : The main ESAWAAI product variables

As a reminder, in the previous table, the resolutions of 17 km, 2 km, and 1 km correspond to the intraburst tile, interburst tile, and OWI product resolutions, respectively.

4.5 Product main global attributes

Regardless of the netCDF group (intraburst or interburst), the global attributes are specified in Table 3. All attributes from the L1C product are retained as they are (for more details, refer to the L1C product description), and a few additional attributes specific to the ESAWAAI project have been included, as shown in Table 3.

	Attribute	Description	Origin
1	Orbit_pass	"Ascending" or "Descending"	XSARSLC processor (L1B)
2	Short_name	The SAR L1 SLC product followed by with the specific sub-swath	
3	Mean_incidence	The mean incidence within the subswath	
4	Required_tile_width_line		
5	Product	SAR product type (SLC for the ESAWAAI project)	
6	Swath	The SAR acquisition mode	
7	Required_tile_overlap_line	The requested overlap between two successive tiles in the azimuth direction	
8	Required_tile_overlap_sample	The requested overlap between two successive tiles in the range direction	
9	Platform_heading	The Sentinel-1 unit heading	
10	Pols	The considered pol for the product (ie VV and VH pour ESAWAAI project)	
11	Stop_date	The end of acquisition time	
12	Footprint	The SAR sub-swath footprint	
13	Safe	The SAR L1 SLC product name	
14	Multidataset	TO EXPLAIN	
15	Start_date	The start date of SAR product acquisition	
16	Coverage	The total coverage of the SAR subswath	
17	Platform	The name of Sentinel-1 unit	
18	Azimuth_steering_rate		
19	lpf	The ESA IPF version for L1 SAR product	
20	Project_name	ESAWAAI	ESAWAAI processor
21	ESAWAAI_S1_L2_OCN_source	The SAR L2 OCN used to extract SAR wind speed, NRCS, heading, incidence at full resolution.	
22	ESAWAAI_S1_L1_GRDH_source	The SAR GRDH product used to infer the rain filter	
23	ESAWAAI_Legacy_dataset_version	The ESAWAAI legacy dataset version	

24	ESAWAAI_Legacy_dataset_aggregation_processor	The processor used to create the legacy dataset of ESAWAAI	
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Table 3: ESAWAAI product attributs description

5 Conclusion

This document provides a comprehensive overview of the legacy dataset for the ESAWAAI project, detailing its structure, content, and key attributes such as nomenclature, resolutions, grids, and other essential metadata. It serves as a reference guide for understanding and utilizing the dataset, offering clear explanations on how to interpret and work with the data. The document will be continuously updated by the data provider throughout the project, ensuring that the dataset remains current and accurately reflects any modifications or additions. This ongoing maintenance makes the document an essential resource for external users, facilitating seamless access and efficient use of the dataset in various research and application contexts.

6 Appendix A : scientific reference

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